

MSc in Computational Software Techniques in Engineering

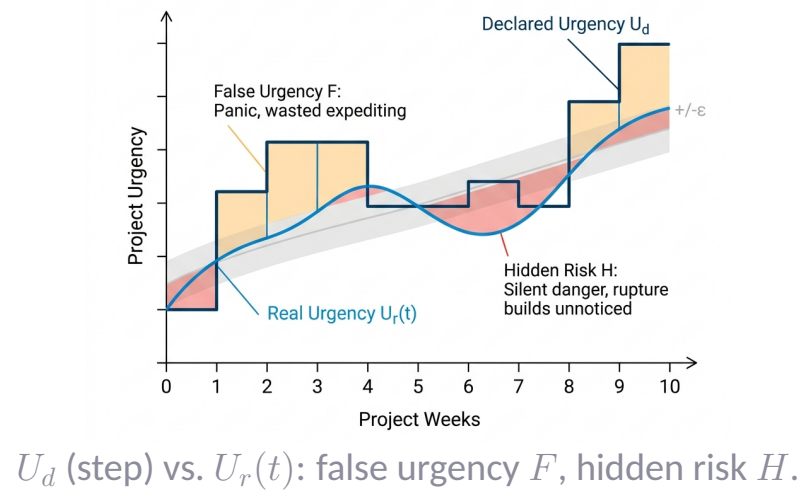
Adequacy Score Between Declared and Real Urgency in a Digital Smart Green Supply Chain

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1. Context & Problem

The **DISCO digital twin** is one of the two bricks of ALLEN-DIN's *Smart Green Supply Chain* programme: a serious game scoring 22 suppliers on cost, delay and CO₂.

Supply chains moving towards **Physical Internet** operation face a signal-quality gap — the urgency an operator *declares* diverges from the urgency that is *physically computable* from the twin.



U_d (step) vs. $U_r(t)$: false urgency F , hidden risk H .

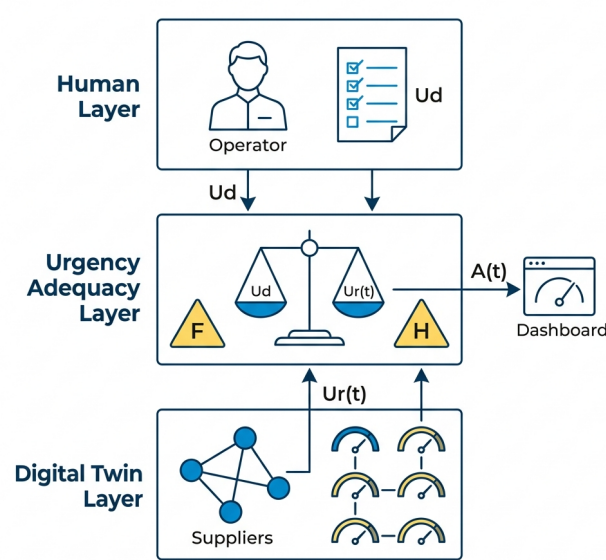
The two pathologies are **not symmetric**: a coordinator who panics wastes premium capacity; one who under-reports misses the window to prevent a line stoppage.

2. Objectives

- A multi-block, DAG-propagated real urgency $U_r(t)$ and a cognitive adequacy score $A(t) \in [0, 100]$.
- A network propagation and criticality layer (PROMETHEE II + systematic shock simulation) exposing structurally critical nodes.
- Empirical validation of the prioritisation logic against recorded operational outcomes.

Six **pre-registered, falsifiable hypotheses** (H1–H6) fix every numeric threshold *before* any data exists: latency, early detection, criticality, predictive validity, contagion, hysteresis.

3. Method: a three-layer model



Real urgency — noisy-OR over six KPI blocks (time, capacity, performance, adverse-event risk, cost, environment), propagated on the supply-chain DAG (β upstream amplification, γ downstream damping):

$$U_{r,loc} = 1 - \prod_m (1 - u_m)^{\omega_m}$$

$$U_{r,i} = 1 - (1 - U_{r,loc,i}) \prod_{j \in \text{Pred}(i)} (1 - \beta_{ji} U_{r,j})$$

Declared urgency — pairwise elicitation over four expert criteria: **AHP** with a consistency-ratio gate, **Fuzzy Best-Worst Method**.

Adequacy — Prospect-Theory asymmetric loss ($\lambda_{\text{under}} = 2.25$, $\lambda_{\text{over}} = 1$, $\alpha = 0.88$):

$$F = [U_d - U_r]_+, \quad H = [U_r - U_d]_+, \quad p = \lambda_{\text{under}} H^\alpha + \lambda_{\text{over}} F^\alpha$$

$$A = 100 \left(e^{-p} - e^{-p_{\text{max}}} \right) / \left(1 - e^{-p_{\text{max}}} \right)$$

Criticality — each node is knocked out in turn; the drop in the final customer's urgency ranks the network (PROMETHEE II net flow ϕ).

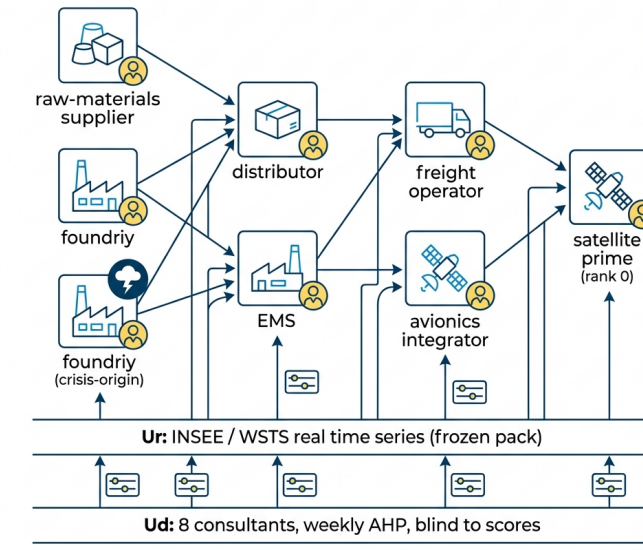
4. Delivered system: SupplyScore

A persistent, **multi-tenant digital twin** in Python on a weekly review cadence.

- Single-facade architecture, gated writes, versioned SQLite (WAL, integrity-checked backups).
- Calibrated event engine — spikes, accidents, transport disruptions, shortages — with weekly Bayesian revision of failure probability.
- Exact additive explainability**: every point of U_d and U_r traced to its block or its upstream arc.
- Twelve-worksheet Excel export: analysis runs outside the tool on frozen data.

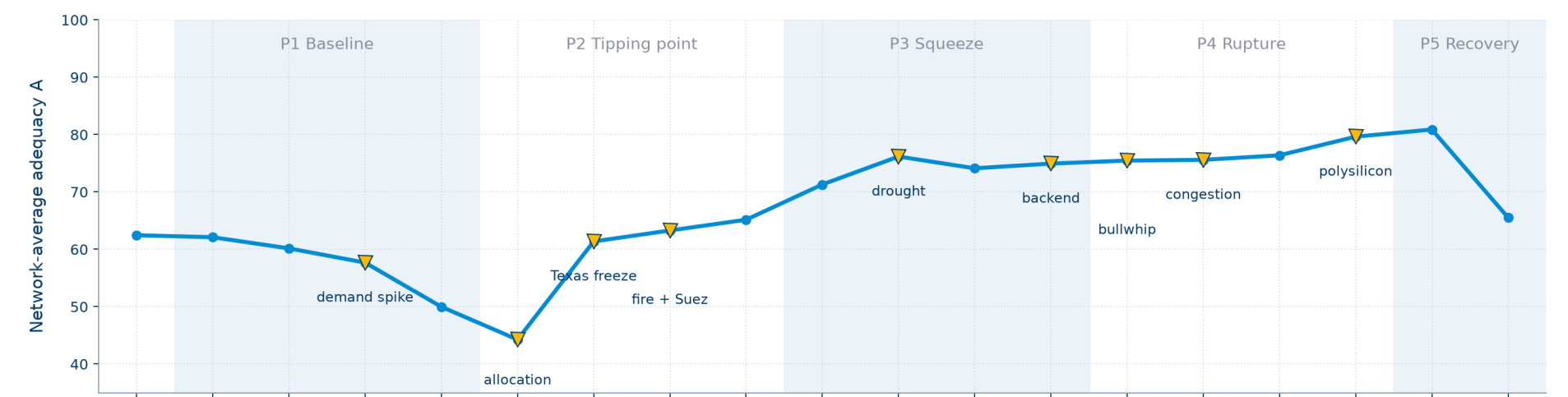
5. Validation: the HÉLIOS serious game

The **2020–2022 semiconductor shortage** replayed as an 8-node, 18-turn serious game over nine working days. U_r is fed by public series (INSEE, WSTS) frozen under SHA-256; U_d comes from the players' weekly AHP questionnaire. Protocol frozen before turn 1, with placebo sheets and a weak-signal turn as controls.



The eight HÉLIOS nodes and the public series feeding U_r .

A **synthetic dry run** (5 iterations) calibrated the null baseline (precision 0.16 / recall 0.27); an exploratory **pilot with eight isolated LLM agents** then ran the full production pipeline.



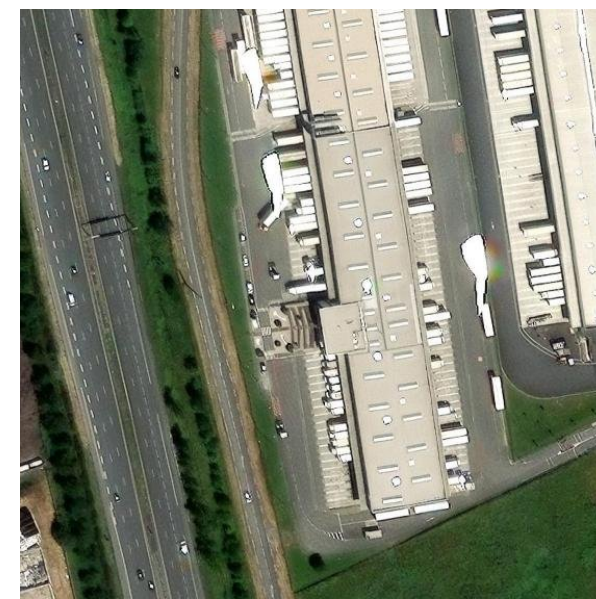
Network-average adequacy A across the 18 turns of the LLM pilot, annotated with the calibrated crisis events (five acts: baseline, tipping point, squeeze, rupture, recovery).

Hypothesis	Verdict	Measured
H1 Cognitive latency	Not confirmed	3 of 8 nodes (thr. 5)
H2 Early detection	Not confirmed	$H_{\text{max}} = 0.017$ (thr. 0.3)
H3 Structural criticality	Not confirmed	top-1 on 70% (thr. 80%)
H4 Predictive validity	Not confirmed	precision / recall 0.00
H5 Urgency contagion	Confirmed	+0.079 vs. +0.005
H6 Perception hysteresis	Not applicable	no measurable decline

Key result. Urgency contagion is reproduced at network scale: nodes *not* physically affected still raise their declared urgency on press-event turns. The LLM readers track their briefing too faithfully to leave the gap H is built to catch — which is why the **human campaign is the only fair test of H4**.

6. Second brick: warehouse volumetry at scale

The Physical Internet needs to know *where* storage capacity sits, of *what type* and *what volume* — at a scale no manual survey reaches. This brick characterises any warehouse from its **GPS coordinates alone**.



- ESRI satellite tiles, zoom 18 (≈ 0.6 m/px), 3×3 grid.
- Ground resolution **calibrated** on the OSM footprint and the 2.40 m standard truck width.
- SAM 3** text-prompted segmentation: *warehouse, loading dock, truck, office*.
- Shape class I / L / U from solidity and compactness.
- Storage volume = footprint $\times$ height (OSM height, or levels $\times 3.5$ m).

One JSON record per site — surface, docks, trucks, shape, volume — i.e. **indicators that aggregate into a national storage-capacity map** and feed the twin with *observed* capacity instead of declared capacity.

7. Conclusions & next steps

A multi-block real urgency, an asymmetric cognitive adequacy score and an operable digital twin were delivered; the measurement apparatus around them is frozen and pre-registered.

- Run the human HÉLIOS campaign — the only run able to decide H1–H6.
- Recalibrate (λ_{under} , λ_{over} , α) on real coordinators, not on literature defaults.
- Connect an ERP feed, so score freshness stops depending on manual entry.
- Fuse the two bricks: observed volumetry as a physical input to the capacity block of U_r .